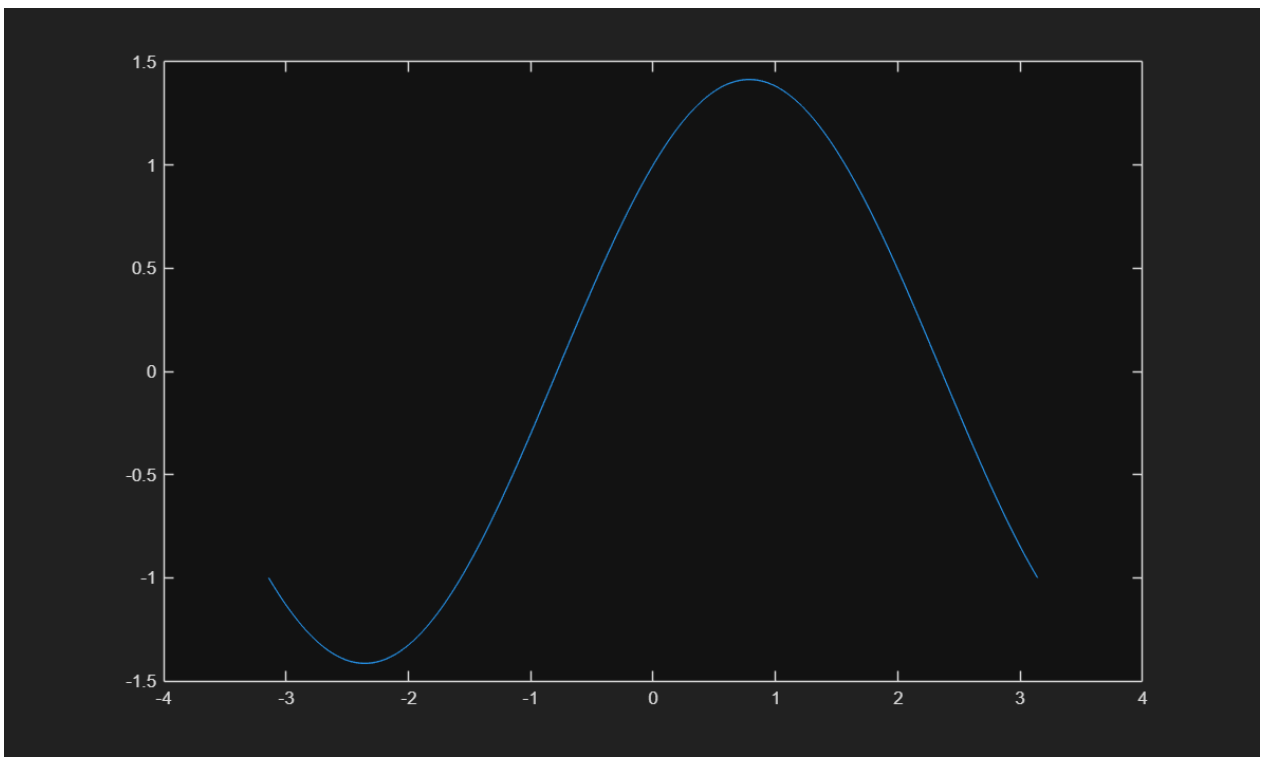
# Lesson 4 - Plotting

Getting matrices and vectors is nice and all, but it would be much nicer to actually visualize what you are calculating. So, there are some ways to make some plots in MATLAB (in fact a lot of ways).

## Some Examples

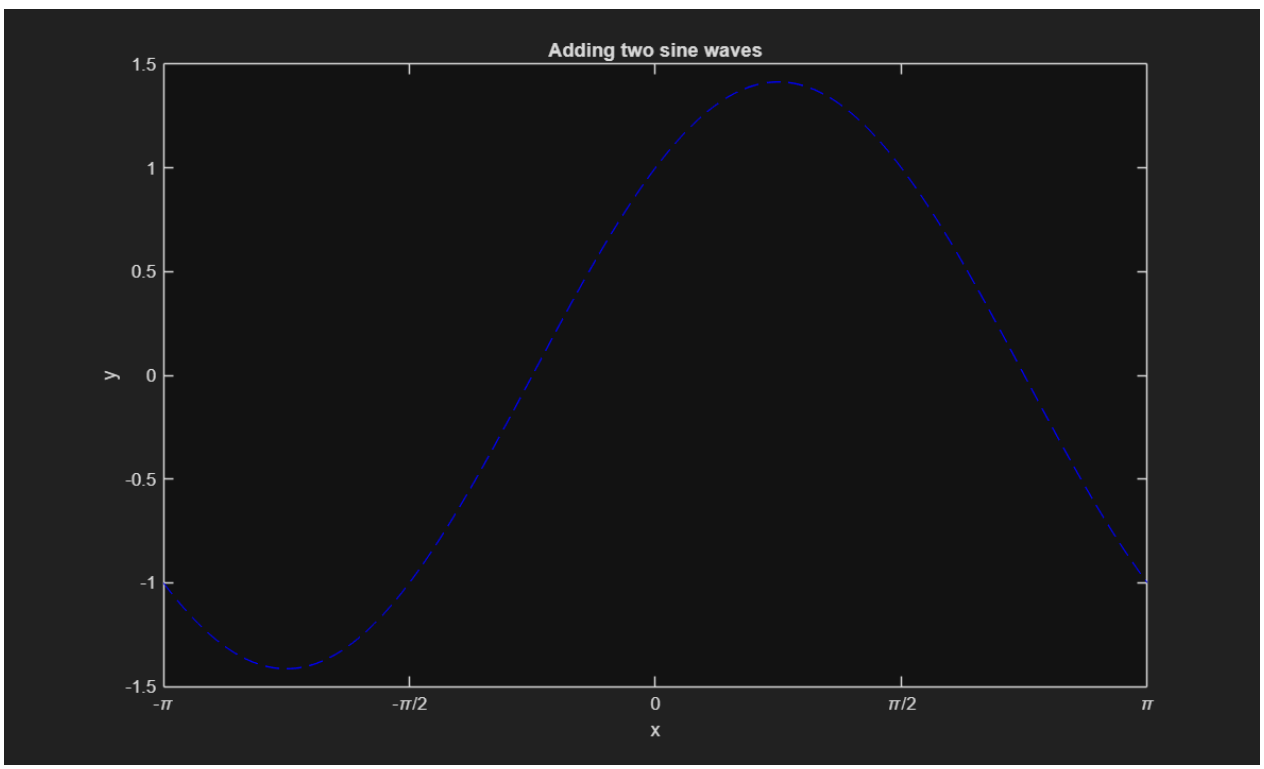
There isn't much to explain with plots, so its better to just jump straight into examples. If we wanted to plot the graph of two trig functions being added together, we might throw some code together like this.

```
x = linspace(-pi, pi, 1000);
y = sin(x) + cos(x);

% figure makes a new figure to plot on (go figure)
figure;
plot(x, y);
```



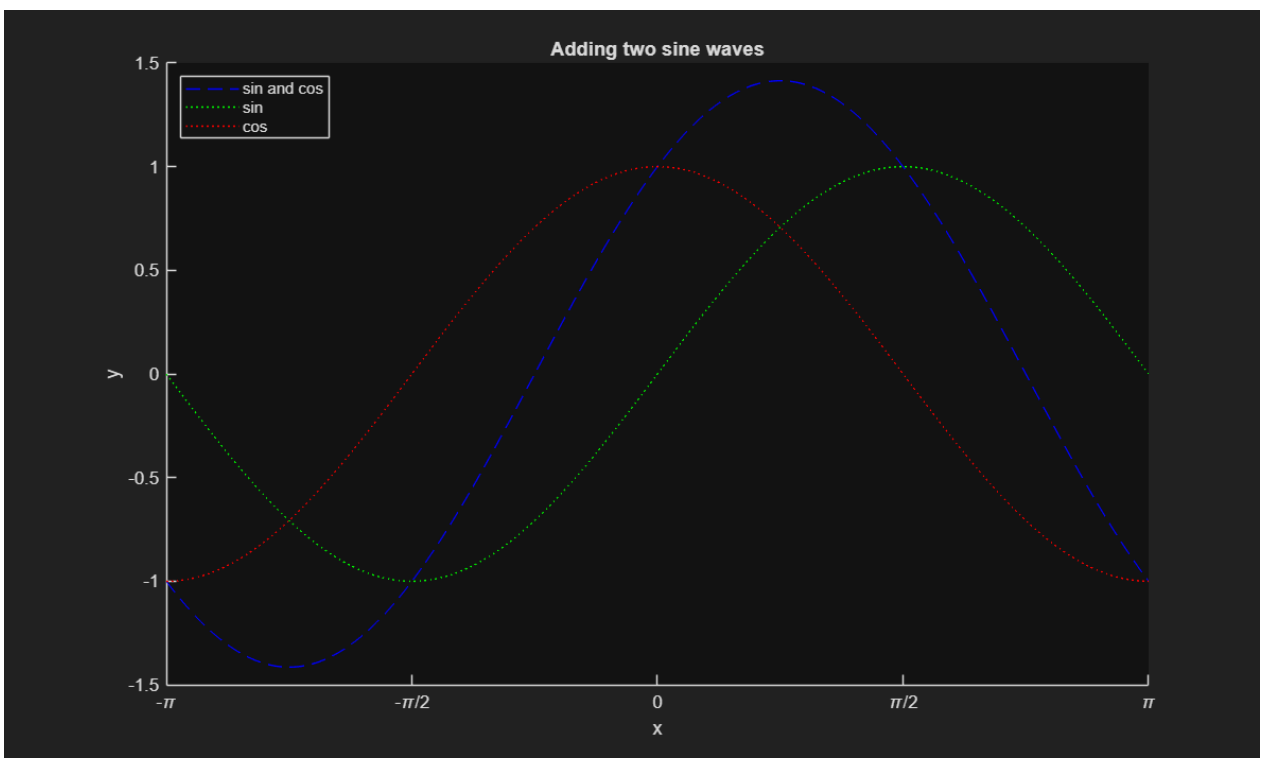
As expected, the figure command creates a figure that we can use plot() on. This certainly plots the added sine waves, but it doesn't look that great. Let's make it look a little prettier at least.

```
x = linspace(-pi, pi, 1000);
y = sin(x) + cos(x);

figure;
plot(x, y, 'b--'); % <- You can change the font style with plot
xlim([x(1), x(end)]);
% Anchors the left and right sides of the plot to these points
xticks([-pi, -pi/2, 0, pi/2, pi]);
% Where to place ticks on x axis
xticklabels({'-\pi', '-\pi/2', '0', '\pi/2', '\pi'});
% What to put on the ticks
title('Adding two sine waves');
% Adds a title
xlabel('x');
% Labels x axis
ylabel('y');
% Labels y axis
```



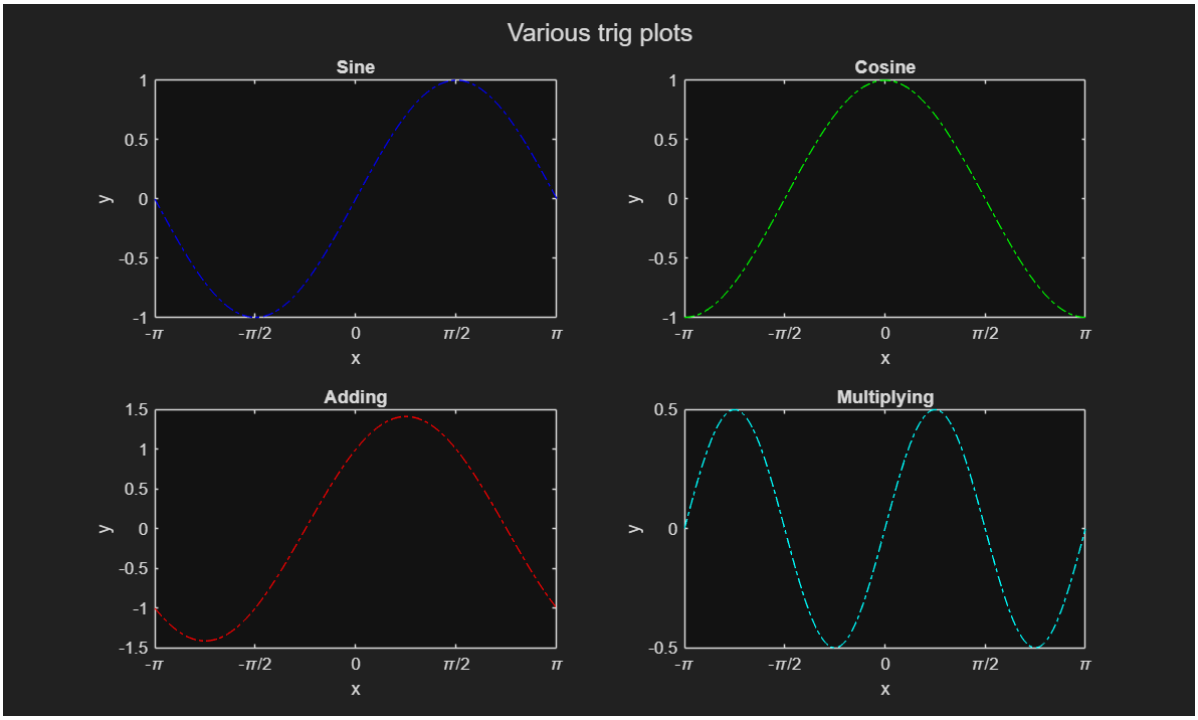
Now, it looks better, but if I try to use plot again, the previous plot gets overwritten. We can try using figure again, but it just makes another plot.

```
x = linspace(-pi, pi, 1000);
y = sin(x) + cos(x);

figure;
% hold on allows you to plot multiple graphs on one figure
hold on;
plot(x, y, 'b--');
plot(x, sin(x), 'g:');
plot(x, cos(x), 'r:');
hold off;

xlim([x(1), x(end)]);
xticks([-pi, -pi/2, 0, pi/2, pi]);
xticklabels({'-\pi', '-\pi/2', '0', '\pi/2', '\pi'});
title('Adding two sine waves');
xlabel('x');
ylabel('y');
legend('sin and cos', 'sin', 'cos', ...
        'Location', 'northwest');
% Creates a legend (the ... allows you to split
% a long line of code into multiple rows in general)
```



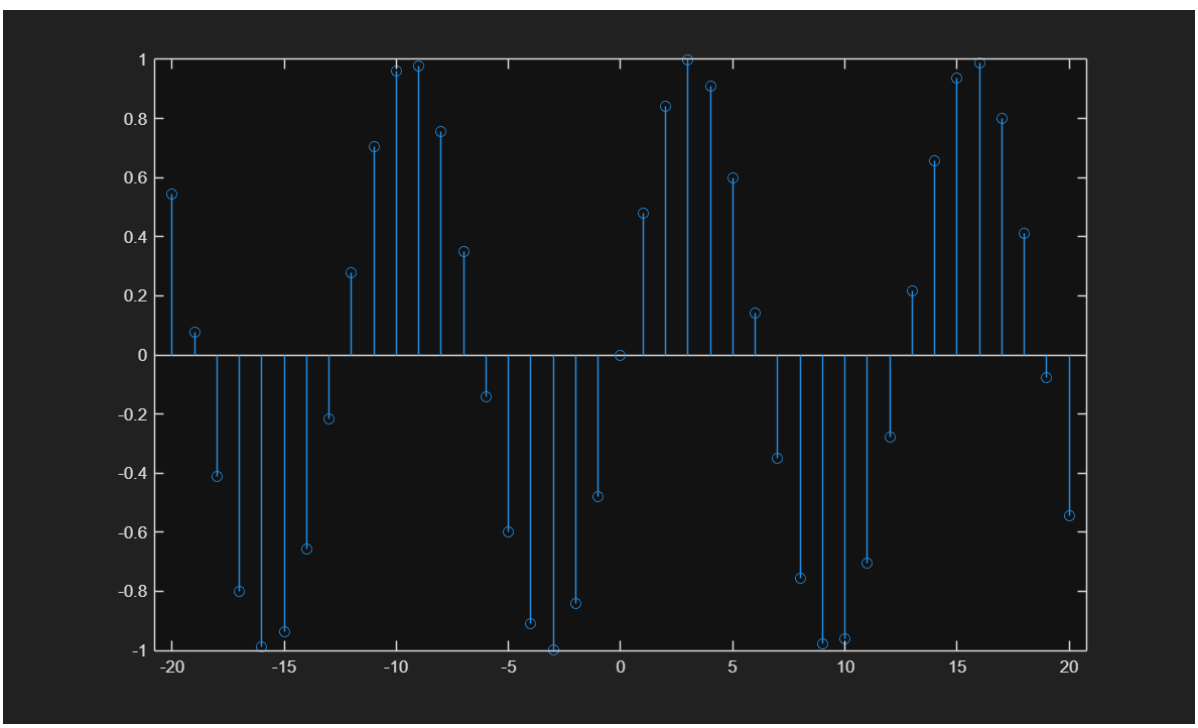
If we want multiple plots within a figure instead, we can use the subplot command.

```
x = linspace(-pi, pi, 1000);
trigsum = sin(x) + cos(x);
trigprod = sin(x) .* cos(x);

figure;
sgtitle('Various trig plots');
% Title for all the plots

% Plot on each individual subplot with subplot (go figure)
subplot(2, 2, 1);
plot(x, sin(x), 'b-.');
xlim([x(1), x(end)]);
xticks([-pi, -pi/2, 0, pi/2, pi]);
xticklabels({'-\pi', '-\pi/2', '0', '\pi/2', '\pi'});
xlabel('x');
ylabel('y');

% For some reason, subplot goes in row-major order for numbering.
subplot(2, 2, 2);
plot(x, cos(x), 'g-.');
xlim([x(1), x(end)]);
xticks([-pi, -pi/2, 0, pi/2, pi]);
xticklabels({'-\pi', '-\pi/2', '0', '\pi/2', '\pi'});
xlabel('x');
ylabel('y');

subplot(2, 2, 3);
plot(x, trigsum, 'r-.');
xlim([x(1), x(end)]);
xticks([-pi, -pi/2, 0, pi/2, pi]);
xticklabels({'-\pi', '-\pi/2', '0', '\pi/2', '\pi'});
xlabel('x');
ylabel('y');

subplot(2, 2, 4);
plot(x, trigprod, 'c-.');
xlim([x(1), x(end)]);
xticks([-pi, -pi/2, 0, pi/2, pi]);
xticklabels({'-\pi', '-\pi/2', '0', '\pi/2', '\pi'});
xlabel('x');
ylabel('y');
```



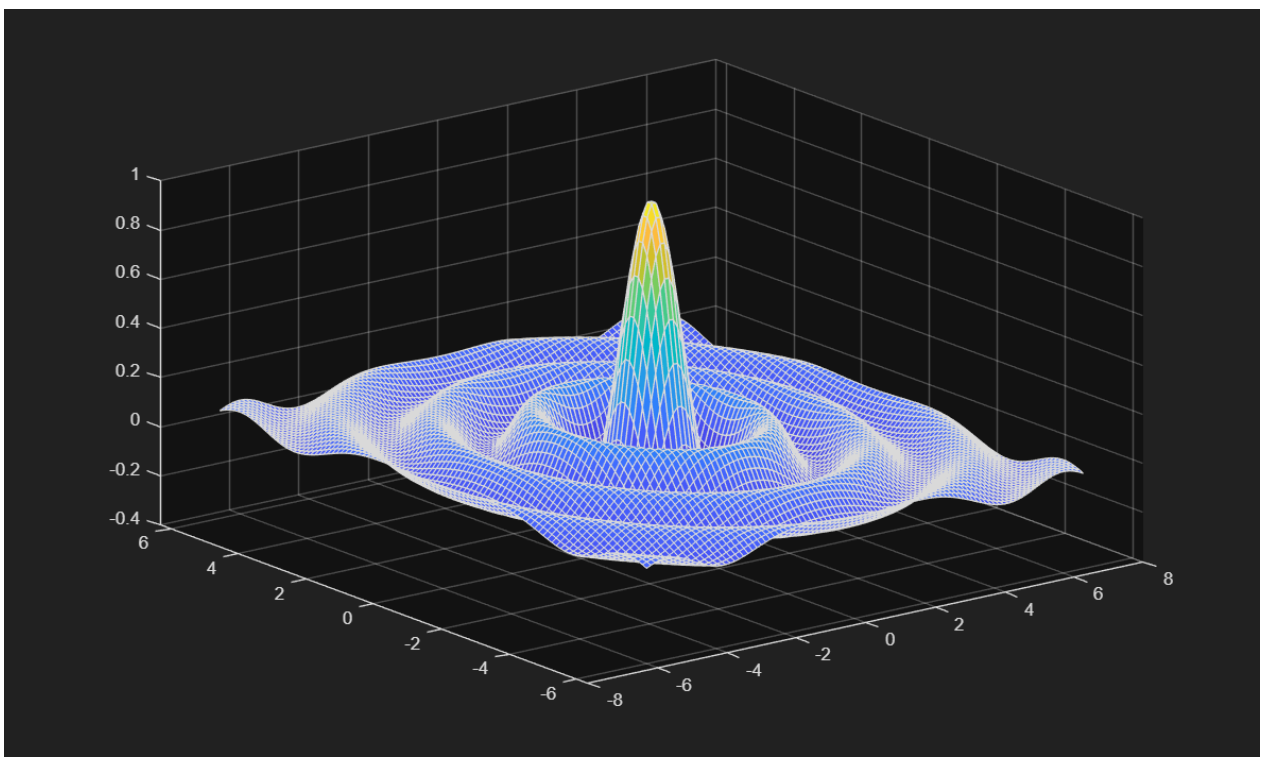

The code for subplots can get repetitive, but you can tidy it up with a for loop in the next lesson. If your data represents discrete data, you can also use stem() for stem plots.

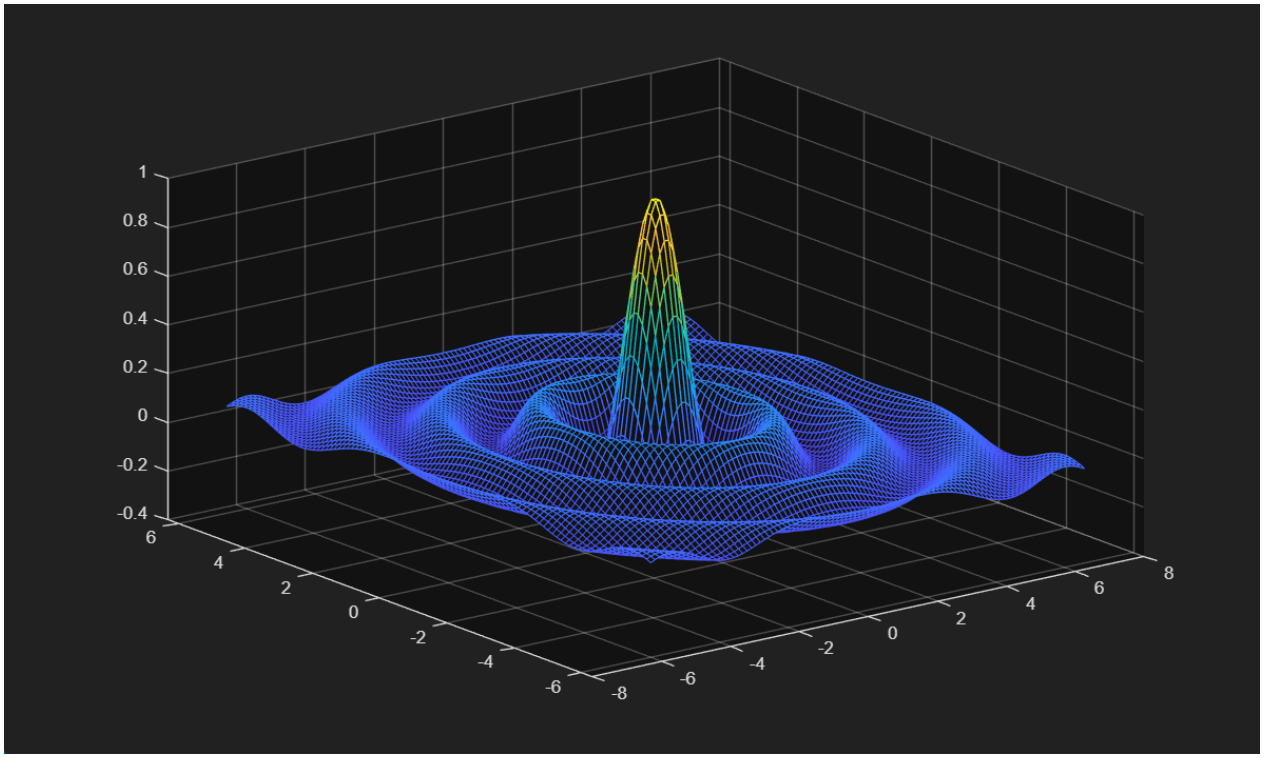
```
n = -20:20;
y = sin(0.5 * n);
figure;
stem(n, y);
```



MATLAB also has 3D plotting if you want to plot functions of two variables.

```
x = linspace(-2*pi, 2*pi, 100);
y = linspace(-2*pi, 2*pi, 100);

[X, Y] = meshgrid(x, y);
% Basically repmat() on x and y to make matrices that show the
% x and y coordinates of all points in the coordinate array

Z = sinc(sqrt(X.^2 + Y.^2));

figure;
surf(X, Y, Z);
% Surface Plot

figure;
mesh(X, Y, Z);
% Mesh Plot
```






# Importing/Exporting Data

Sometimes, you may have to import or export data to use for various projects. After uploading any imported data you need to your MATLAB directory, MATLAB has a *lot* of functions for dealing with importing and exporting data to your scripts.

A common type of format you might import is a .mat file which is the format used for saving and loading MATLAB variables (where as .m files are the actual code/scripts you write). The load() and save() functions help import and export data as .mat files.

Other commonly used commands include readmatrix and writematrix for csv or xsl files, imread and imwrite for images (png, jpg, gif, etc.), and audioread and audiowrite for audio formats (). The full list can be found on `https://www.mathworks.com/help/matlab/import_export/supported-file-formats-for-import-and-export.html` for the many types of formats and functions supported by MATLAB.

# Additional Documentation

This is just a start to the possible plots you can make in MATLAB. There are significantly more types and variations such as stem plots, polar plots, contour plots, histograms, vector fields, and much much more. As there is not enough time to go over all of these different kinds of plots, you can go find out more about these different types of plots at `https://www.mathworks.com/help/matlab/2-and-3d-plots.html`.